Minor Project

AI Personal Health Assistant & Medicine Management Platform

Your everyday companion for health tracking and medicine reminders

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AI Personal Health Assistant & Medicine Management

Abstract

AI Personal Health Assistant & Medicine Management System is a web-based application that helps users manage their personal health information and daily healthcare activities. It provides medicine reminders, health record management, medical report storage, appointment management, and AI-based health assistance. The system aims to provide a centralised and user-friendly platform for managing healthcare information and receiving relevant AI-based health guidance.

Existing Problems

- Maintaining physical medical documents
- Difficulty accessing health information when required
- Forgetting medicine schedules
- Missed appointments
- Scattered medical reports and health records
- Lack of an interactive health-support system

Proposed Solution

- Centralised digital health management
- Medicine and appointment reminders
- Medical report storage and retrieval
- Health record management
- AI-based health assistance
- Secure user and admin management

SCOPE & MAJOR FUNCTIONALITIES

User/Patient

- Registration & Authentication
- Manage Personal Information
- Manage Emergency Contact
- Manage Medical Reports
- Manage Medicines
- Medicine Reminders
- Manage Appointments
- Manage Health Records
- Submit Feedback
- AI Health Assistance

Admin

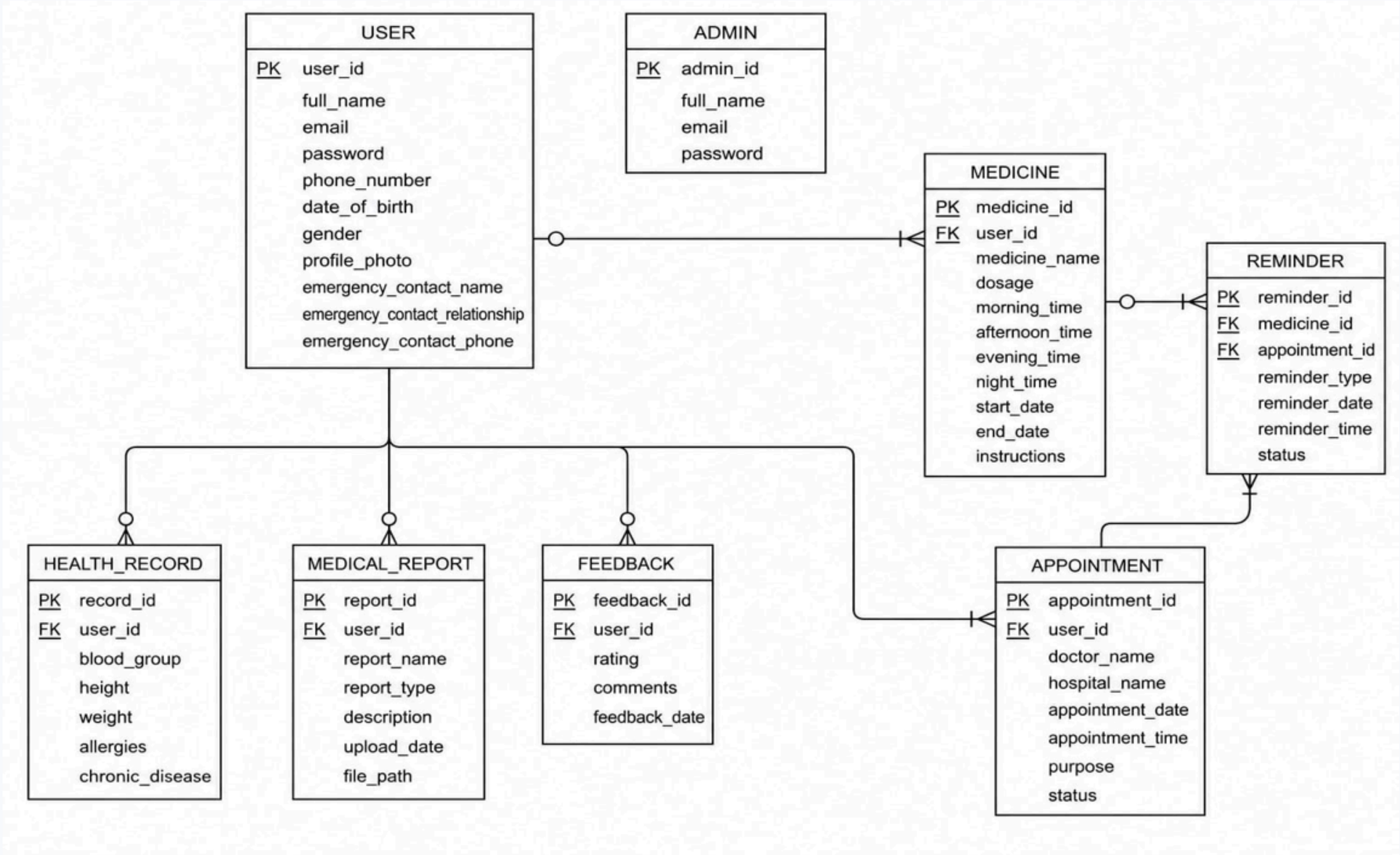
- Admin Authentication
- Manage Users
- Manage Medical Reports
- Manage Medicines
- Manage Appointments
- Manage Health Records
- Manage Feedback

Technology Stack

Technologies used to build the Health Assistant platform

- **Frontend**: HTML5, CSS3, JavaScript, Bootstrap 5
- **Backend**: Python
- **Database**: MySQL
- **AI Integration:** Gemini API

E-R DIAGRAM


USER
PK user_id
full_name
email
password
phone_number
date_of_birth
gender
profile_photo
emergency_contact_name
emergency_contact_relationship
emergency_contact_phone
ADMIN
PK admin_id
full_name
email
password
MEDICINE
PK medicine_id
FK user_id
medicine_name
dosage
morning_time
afternoon_time
evening_time
night_time
start_date
end_date
instructions
REMINDER
PK reminder_id
FK medicine_id
FK appointment_id
reminder_type
reminder_date
reminder_time
status
HEALTH_RECORD
PK record_id
FK user_id
blood_group
height
weight
allergies
chronic_disease
MEDICAL_REPORT
PK report_id
FK user_id
report_name
report_type
description
upload_date
file_path
FEEDBACK
PK feedback_id
FK user_id
rating
comments
feedback_date
APPOINTMENT
PK appointment_id
FK user_id
doctor_name
hospital_name
appointment_date
appointment_time
purpose
status

E-R DIAGRAM

Main Entities

- User
- Admin
- Medicine
- Reminder
- Appointment
- Health Record
- Medical Report
- Feedback

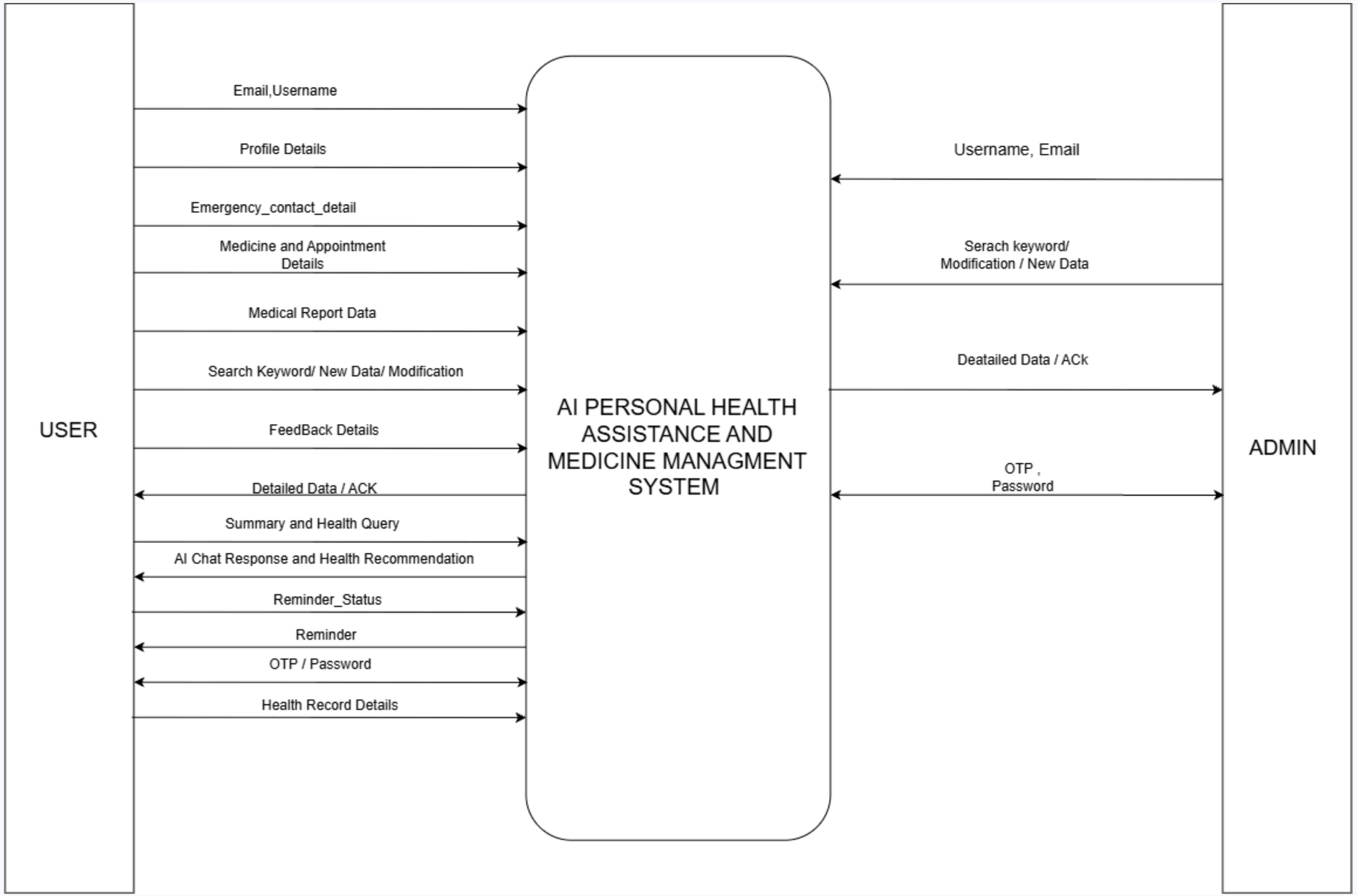
Key Relationships

- User → Medicine 1:M
- User → Appointment 1:M
- User → Medical Report 1:M
- User → Feedback 1:M
- User → Health Record 1:1
- Medicine → Reminder 1:M
- Appointment → Reminder 1:M
- Admin → User 1:M

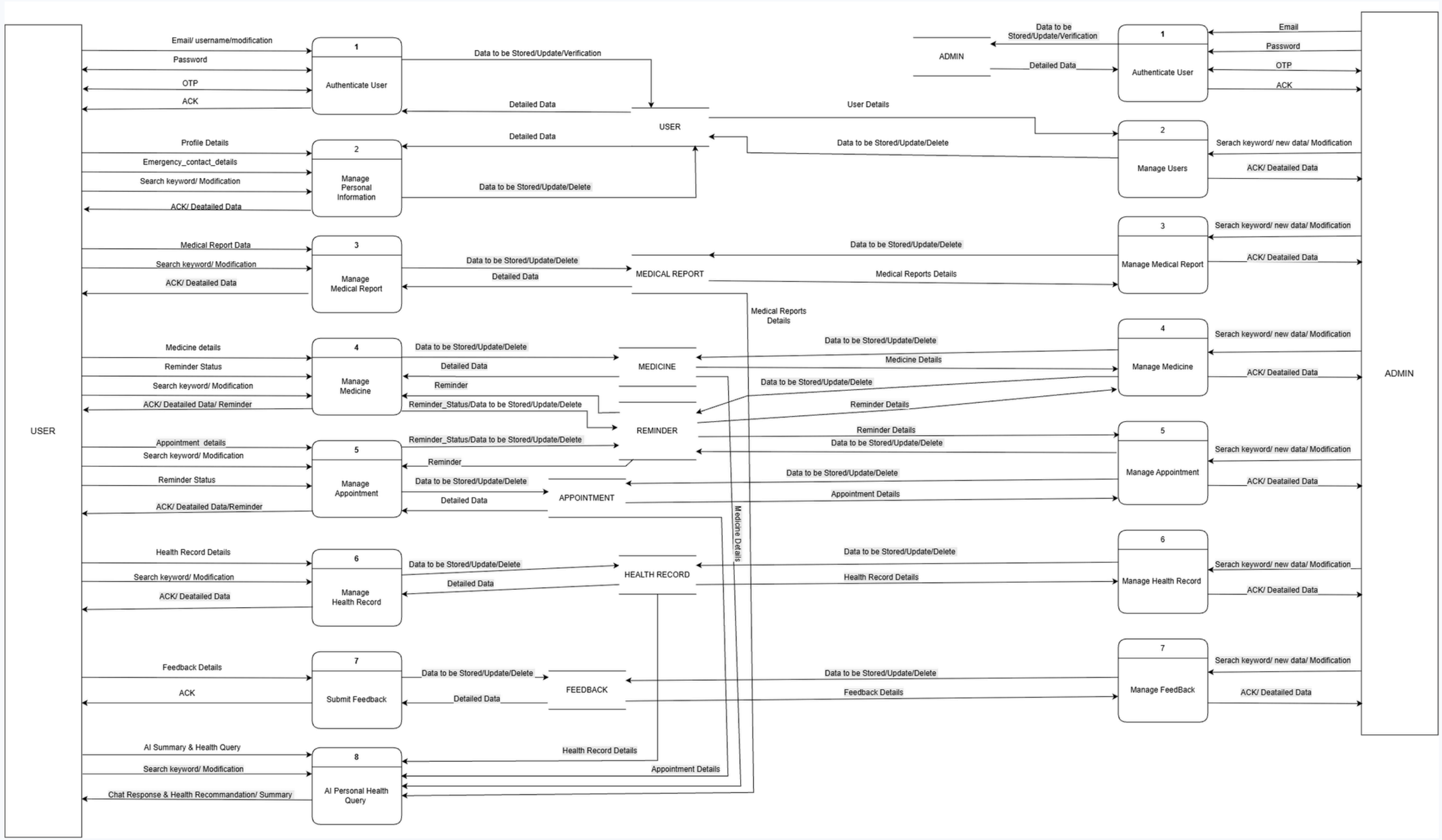
WHO USES IT

- Elderly & chronic-illness patients
- Busy individuals managing multiple medicines
- Caregivers & family members
- General wellness-conscious users

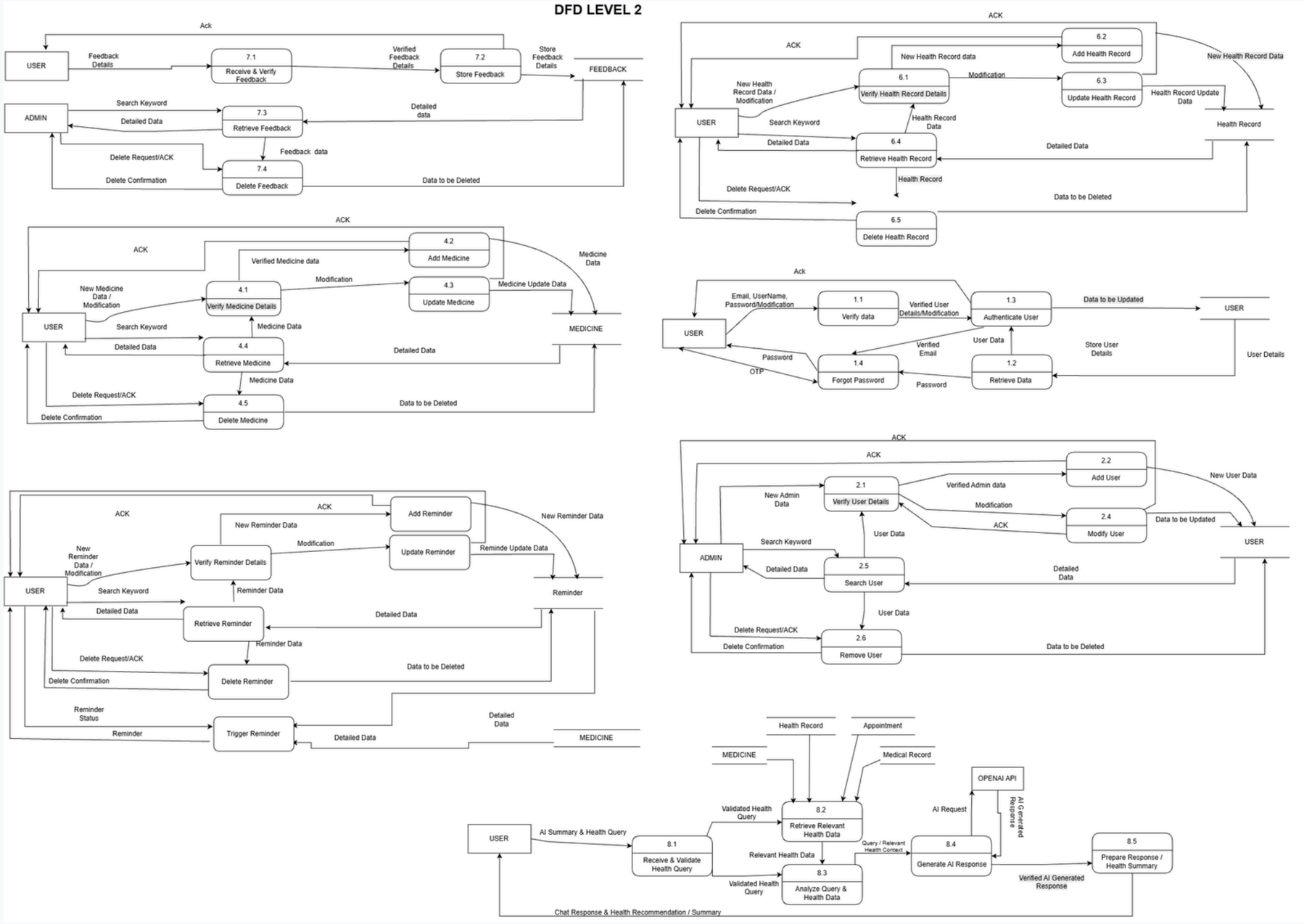
DFD LEVEL 0 — CONTEXT DIAGRAM



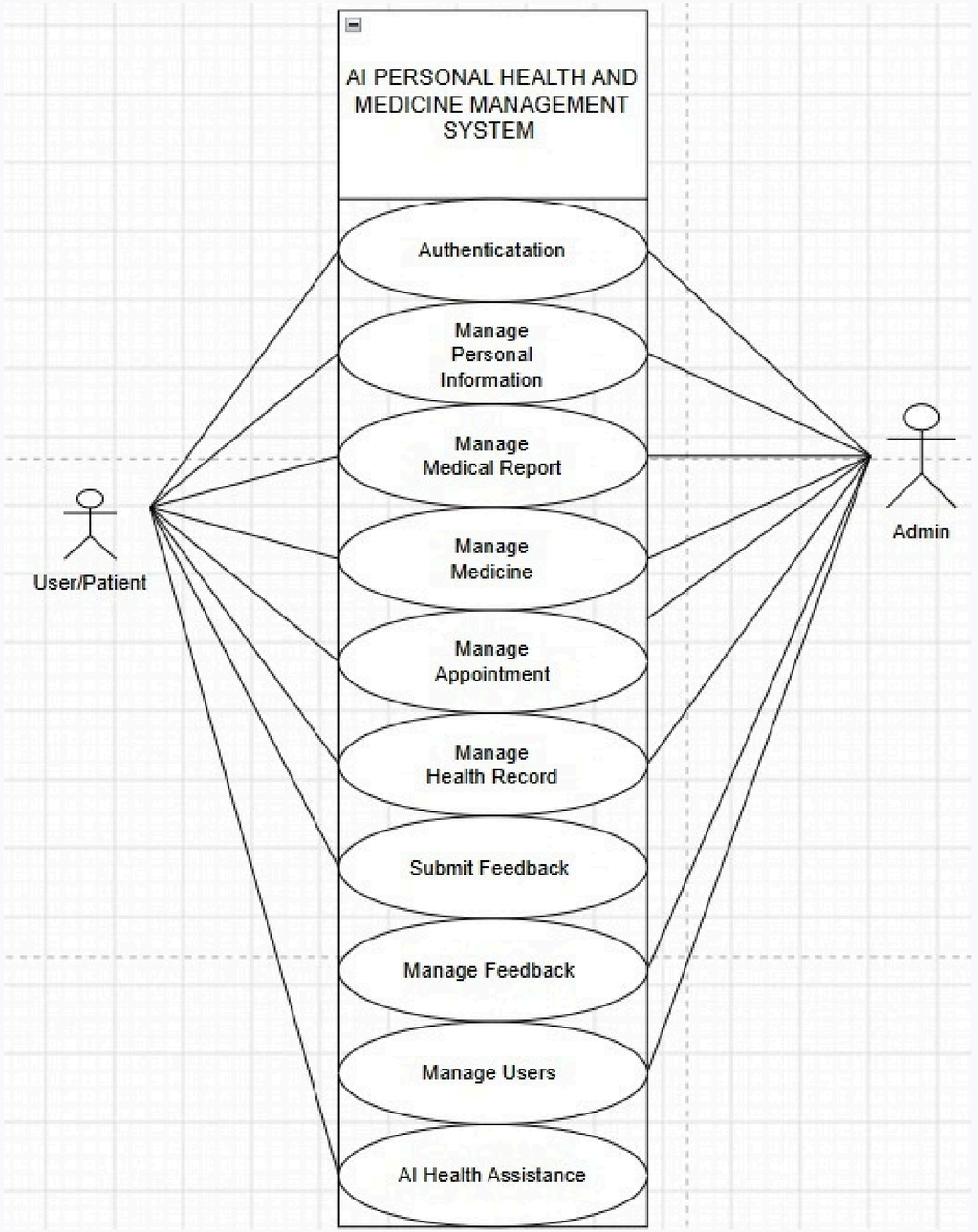
DFD LEVEL 1


USER
Email/ username/modification
Password
OTP
ACK
1
Authenticate User
Data to be Stored/Update/Verification
Detailed Data
USER
Profile Details
Emergency_contact_details
Search keyword/ Modification
ACK/ Deatailed Data
2
Manage Personal Information
Detailed Data
Data to be Stored/Update/Delete
Medical Report Data
Search keyword/ Modification
ACK/ Deatailed Data
3
Manage Medical Report
Data to be Stored/Update/Delete
Detailed Data
MEDICAL REPORT
Medical Reports Details
Medicine details
Reminder Status
Search keyword/ Modification
ACK/ Deatailed Data/ Reminder
4
Manage Medicine
Data to be Stored/Update/Delete
Detailed Data
Reminder
Reminder_Status/Data to be Stored/Update/Delete
MEDICINE
REMINDER
Appointment details
Search keyword/ Modification
Reminder Status
ACK/ Deatailed Data/Reminder
5
Manage Appointment
Reminder_Status/Data to be Stored/Update/Delete
Reminder
Data to be Stored/Update/Delete
Detailed Data
APPOINTMENT
Medicine Details
Health Record Details
Search keyword/ Modification
ACK/ Deatailed Data
6
Manage Health Record
Data to be Stored/Update/Delete
Detailed Data
HEALTH RECORD
Feedback Details
ACK
7
Submit Feedback
Data to be Stored/Update/Delete
Detailed Data
FEEDBACK
AI Summary & Health Query
Search keyword/ Modification
Chat Response & Health Recommandation/ Summary
8
AI Personal Health Query
Health Record Details
Appointment Details
ADMIN
Data to be Stored/Update/Verification
Detailed Data
1
Authenticate User
Email
Password
OTP
ACK
User Details
Data to be Stored/Update/Delete
2
Manage Users
Serach keyword/ new data/ Modification
ACK/ Deatailed Data
Data to be Stored/Update/Delete
Medical Reports Details
3
Manage Medical Report
Serach keyword/ new data/ Modification
ACK/ Deatailed Data
Data to be Stored/Update/Delete
Medicine Details
Data to be Stored/Update/Delete
Reminder Details
4
Manage Medicine
Serach keyword/ new data/ Modification
ACK/ Deatailed Data
Reminder Details
Data to be Stored/Update/Delete
5
Manage Appointment
Serach keyword/ new data/ Modification
ACK/ Deatailed Data
Data to be Stored/Update/Delete
Appointment Details
Data to be Stored/Update/Delete
Health Record Details
6
Manage Health Record
Serach keyword/ new data/ Modification
ACK/ Deatailed Data
Data to be Stored/Update/Delete
Feedback Details
7
Manage FeedBack
Serach keyword/ new data/ Modification
ACK/ Deatailed Data
ADMIN

DFD LEVEL 2


DFD LEVEL 2
USER
ADMIN
FEEDBACK
MEDICINE
7.1
Receive & Verify Feedback
7.2
Store Feedback
7.3
Retrieve Feedback
7.4
Delete Feedback
6.1
Verify Health Record Details
6.2
Add Health Record
6.3
Update Health Record
6.4
Retrieve Health Record
6.5
Delete Health Record
4.1
Verify Medicine Details
4.2
Add Medicine
4.3
Update Medicine
4.4
Retrieve Medicine
4.5
Delete Medicine
1.1
Verify data
1.2
Retrieve Data
1.3
Authenticate User
1.4
Forgot Password
2.1
Verify User Details
2.2
Add User
2.4
Modify User
2.5
Search User
2.6
Remove User
Add Reminder
Update Reminder
Verify Reminder Details
Retrieve Reminder
Delete Reminder
Trigger Reminder
Reminder
OPENAI API
8.1
Receive & Validate Health Query
8.2
Retrieve Relevant Health Data
8.3
Analyze Query & Health Data
8.4
Generate AI Response
8.5
Prepare Response / Health Summary
Health Record
Appointment
Medical Record
AI Summary & Health Query
Validated Health Query
Relevant Health Data
Query / Relevant Health Context
AI Request
AI Generated Response
Verified AI Generated Response
Chat Response & Health Recommendation / Summary

USE CASE DIAGRAM



Thank You!